Estimating Baseflow Recession Constants from Sparse Gauge Records

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Abstract. Baseflow recession constants are difficult to estimate where streamflow gauges report only a few times per month. We present a simple resampling procedure that recovers the recession constant from sparse records. Tested on 41 catchments, the method reduced the median estimation error from 31% to 9% relative to a naive fit.

1. Introduction

The recession of streamflow after rainfall has long been described by an exponential decay of the form $Q(t) = Q_0 \exp(-t/k)$, where k is the recession constant. Reliable values of k are needed to forecast low flows and to size water supply storage.

In many regions, however, gauges are read manually and the record contains only a handful of values each month. Standard fitting procedures assume daily data and give biased estimates when this assumption fails.

This paper makes two contributions. First, we quantify the bias of the naive fit as a function of sampling interval. Second, we propose a resampling correction that needs no additional measurements.

Earlier work has treated this problem either by discarding sparse records altogether or by interpolating them to a daily grid. Interpolation, however, smooths the very curvature that the recession constant is meant to describe.

The remainder of the paper is organised as follows. Section 2 describes the data and the method, Section 3 reports the results, and Sections 4 and 5 discuss their limits and draw conclusions.

2. Data and methods

2.1 Study catchments

We used 41 catchments with at least ten years of daily discharge, drawn from three climatic regions. Their main characteristics are summarised in Table 1.

Daily series were thinned to emulate sparse reporting. Intervals of 3, 7 and 15 days were generated by retaining every n-th observation[1], and each experiment was repeated with 200 random starting offsets.

2.2 Resampling correction

The correction fits the recession constant to many random subsets of the thinned record and then adjusts the median fit with a bias factor calibrated on the daily series.

2.3 Evaluation

Performance is measured by the relative error of the fitted recession constant against the value obtained from the full daily series. We report the median over all catchments and offsets, because the distribution of errors is strongly skewed.

Table 1. Characteristics of the study catchments.

Region	Catchments	Median area (km²)	Mean k (days)
Humid temperate	17	212	38.4
Semi-arid	14	486	61.2
Alpine	10	95	24.7

3. Results

With daily data the naive and corrected fits agree closely. As the sampling interval grows, the naive fit overestimates the recession constant, because slow late-recession flows are over-represented among the retained points.

The corrected estimate stays within 10% of the daily-data value for all intervals up to 15 days. Table 2 gives the median relative error for each interval and region.

The spread across offsets also shrinks after correction. For the 15-day interval the interquartile range of the error fell from 24 to 7 percentage points, indicating that the correction removes variance as well as bias.

Sensitivity to the number of random subsets was small: beyond 100 subsets the median estimate changed by less than one percent, so the value of 200 used here is conservative.

Table 2. Median relative error of the recession constant.

Interval (days)	Naive (%)	Corrected (%)
3	8	2
7	19	5
15	31	9

Errors were largest in the semi-arid region, where long dry spells leave few usable recession segments. In the alpine region snowmelt interrupted many recessions, and fewer segments were available for the fit.

4. Discussion

The improvement is not limited to a single region, which suggests that the bias is a property of the sampling rather than of the catchments.

A practical implication is that existing manual gauge networks need not be replaced to support low-flow forecasting. Where funds allow, a modest increase in reading frequency during dry seasons would add further value.

We also note that the bias factor varies smoothly with the sampling interval, so values for intervals not tested here can be interpolated with reasonable confidence.

Two limitations deserve mention. The correction assumes a single linear reservoir and may be less accurate where several reservoirs contribute to the recession. It also needs at least five years of record to calibrate the bias factor.

5. Conclusions

Sparse gauge records can support reliable recession analysis if the sampling bias is corrected. The procedure is easy to implement and we provide a reference script with the supplementary material.

Data availability

Discharge records are available from the regional authorities on request. The resampling script and the calibrated bias factors are included in the supplementary material.

Acknowledgements

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[1] Observations were retained at fixed intervals after discarding days with missing values.

References

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